A LOCALLY FINITE TYPE MORPHISM SENDS FINITE TYPE POINTS TO FINITE TYPE POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is three lines and is complete: it produces the test morphism, composes, and reads off the definition.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The definition of a finite type point of an algebraic space. used at proof lines 5099-5103. Fixes what has to be shown for f(x).
- Permanence of local finite typeness under composition. Lemma lemma-composition-finite-type, used at proof lines 5101-5102. Makes the composite Spec(k) -> Y locally of finite type.
- Representation of a point by a locally finite type morphism from a field. used at proof lines 5100-5101. Supplies the test morphism for x.

Source reference. Original lemma 03XG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5147–5147. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is locally of finite type, then $f(X_{ft\text{-}pts}) \subset Y_{ft\text{-}pts}$.*

Proof. Take $x \in X_{\text{ft-pts}}$. Represent x by a locally finite type morphism $x : \text{Spec}(k) \to X$. Then $f \circ x$ is locally of finite type by Lemma 03XG. Hence $f(x) \in Y_{\text{ft-pts}}$. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.